COMPUTATIONAL DYNAMICS - LECTURE 10

Articulated Body Inertias

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Articulated Body Inertias - Overview

- Recap
- Inter-body spatial force decompositions
  - Terminal-body model
  - Composite-body model
  - Articulated-body model
- Single body equations of motion decomposition
- Floppy Articulated body model
- Properties of the Articulated Body Inertia
- ATBI inertia for hinge special cases
- Non-Floppy Articulated body model
- ATBI residual for hinge special cases
- Articulated body model
- Connections to Estimation Theory

Recap

- Newton-Euler factorization of the mass matrix
- Introduced composite rigid body (CRB) inertias for the decomposition of the mass matrix and its $O(N^2)$ computation instead of $O(N^3)$ computation
- Developed operator form of system equations of motion
- Developed O(N) Newton-Euler inverse dynamics algorithm
- Explored inverse dynamics based computation of mass matrix, and CRB based inverse dynamics and force decompositions

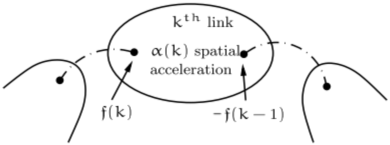
Background

- Now switch to the **forward dynamics**
  - Involves the mass matrix inverse
- Taking a step back to work at the component level to build up some physical intuition
- Could go directly to the spatial operator route
- This route will involve a new quantity referred to as the articulated body inertia

Inter-body spatial force decompositions

Inter-body spatial force decompositions - models

- **Terminal**, **Composite Rigid Body**, and **Articulated** models
- Ignore Coriolis terms for the moment



Terminal-body model $\quad \mathfrak{f}(k) = \boxed{M(k)\,\alpha(k)} + \boxed{\phi(k,k-1)\mathfrak{f}(k-1)}$

Composite-body model $\quad \mathfrak{f}(k) = \boxed{\mathcal{R}(k)\,\alpha(k)} + \boxed{y(k)}$

Articulated-body model $\quad \mathfrak{f}(k) = \underbrace{\boxed{\mathcal{P}(k)\,\alpha(k)}}_{\text{inertia term}} + \underbrace{\boxed{\xi(k)}}_{\text{Residual force term}}$

Inter-body spatial force decompositions

- Ignore Coriolis terms for the moment as they only add notational clutter for now
- Force decompositions consists of inertia + residual terms
- From the equations of motion (**Terminal**) we had

$$\mathfrak{f}(k) = \underbrace{\boxed{M(k)\ \alpha(k)}}_{\text{depends on the kth body}} + \underbrace{\boxed{\phi(k, k-1)\ \mathfrak{f}(k-1)}}_{\text{depends on all bodies}}$$

- Not a very useful decomposition but its a good place to start

Inter-body spatial force decompositions

- Ignore Coriolis terms for the moment as they only add notational clutter for now
- Force decompositions consists of inertia + residual terms
- From the equations of motion (**Composite**/CRBs) we had

$$\mathfrak{f}(k) = \underbrace{\boxed{R(k)\ \alpha(k)}}_{\text{depends on outboard body}} + \underbrace{\boxed{y(k)}}_{\text{depends on outboard generalised acclerations}}$$

- The more complex inertia term simplifies the residual force term in the force decompositions

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Inter-body spatial force decompositions - Why force decomposition?

- Force decompositions provides a way to view the dynamics in a different way
  - View the general multibody system as a deviation from a reference model
- Composite body inertias provide an alternative way to describe the acceleration force relationship and provide additional insight
- The decompositions also provide a pathway to computational algorithms, eg. the CRBs based inverse dynamics algorithms we have seen previously
- The articulated body force decomposition will provide the basis for the $O(N)$ recursive forward dynamics algorithm

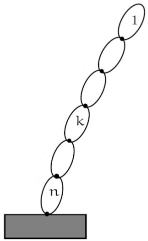
Single body equations of motion decomposition

Single body equations of motion decomposition - Terminal body model

- For the terminal body model the k^{th} body is viewed as a terminal body, meaning the connections to the child bodies is ignored initially.
- This means that the k^{th} body only feels forces from its parent, $\mathfrak{f}(k)$.
- Residual term is zero when the k^{th} body is a terminal body.
- Force acceleration relationship

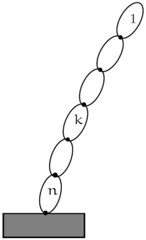
$$\mathfrak{f}(k) = M(k)\ \alpha(k)$$

- This is only true for bodies that are truly terminal bodies



Single body equations of motion decomposition - Terminal body model

- For the terminal body model the k^{th} body is viewed as a terminal body, meaning the connections to the child bodies is ignored initially.
- This means that the k^{th} body only feels forces from its parent, $\mathfrak{f}(k)$.
- Residual term is zero when the k^{th} body is a terminal body
- In reality the body is non-terminal, there are outboard bodies, and the residual term accounts for the interaction with all the outboard bodies

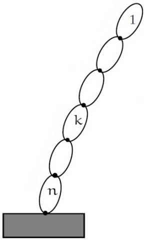


$$\mathfrak{f}(k) = M(k)\,\alpha(k) + \phi(k, k-1)\,\mathfrak{f}(k-1)$$

Single body equations of motion decomposition - Composite body model

- An improvement over terminal body model as outboard bodies are not ignored
- In this model all outboard bodies are assumed to be locked in all DoF
- Residual term is zero if outboard generalized accelerations are zero as outboard bodies are assumed rigid

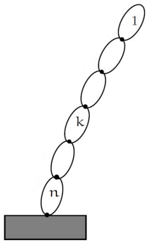
$$\mathfrak{f}(k) = R(k)\,\alpha(k)$$



Single body equations of motion decomposition - Composite body model

- In reality the outboard bodies are not locked and have non-zero generalized accelerations
- The residual term accounts for the non-zero generalized accelerations of the outboard bodies
  - The generalised accelerations of the inboard bodies does not matter

$$\mathfrak{f}(k) = R(k)\,\alpha(k) + y(k)$$



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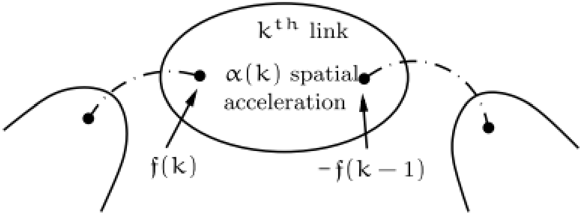
Floppy Articulated body model

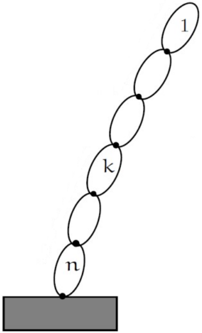
Floppy Articulated body model



Terminal-body model	$\mathfrak{f}(k) = M(k)\,\alpha(k) + \phi(k, k-1)\mathfrak{f}(k-1)$
Composite-body model	$\mathfrak{f}(k) = \mathcal{R}(k)\,\alpha(k) + y(k)$
Articulated-body model	$\mathfrak{f}(k) = \mathcal{P}(k)\,\alpha(k) + \xi(k)$

Floppy Articulated body model

- In the "floppy" case, the outboard hinges are free with zero generalized forces.
- Later non-zero generalized forces will be allowed
- What is the effective inertia (the force/acceleration relationship) at the $(k+1)^{th}$ body when the outboard bodies are floppy (the outboard body generalized forces are all 0)?



Tip body’s articulated body inertia

- The tip body has no children to account for, so the only force is from the parent

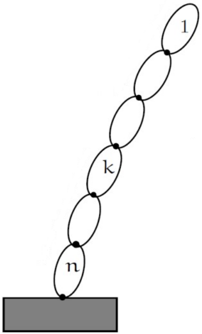
$$\mathfrak{f}(1) = M(1)\,\alpha(1)$$

$$\boxed{P(1) = M(1)}$$

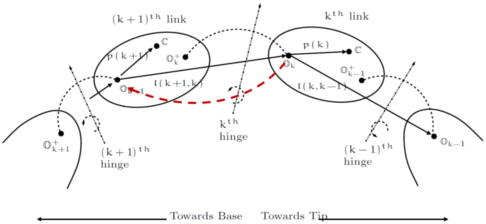
Articulated body inertia

- Will use induction based argument to extend to other bodies
- Let us assume we have established the articulated body inertia (ATBI) relationship for the k^{th} body

$$\mathfrak{f}(k) = P(k)\,\alpha(k)$$



Floppy Articulated body model - Induction based derivation



- Assume we know the relationship for the k^{th} body:

$$\mathfrak{f}(k) = P(k)\ \alpha(k)$$
$$\tau(j) = H(j)\ \mathfrak{f}(j) = 0 \quad \forall\ j < k$$

- Want to establish the relationship for body $(k+1)^{th}$

$$\mathfrak{f}(k+1) = P(k+1)\ \alpha(k+1)$$
$$\tau(j) = H(j)\ \mathfrak{f}(j) = 0 \quad \forall\ j < k+1$$

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Floppy Articulated body model - Conditions to be met

- Assumptions

$$\mathfrak{f}(k) = P(k)\, \alpha(k)$$
$$\tau(j) = H(j)\, \mathfrak{f}(j) = 0 \quad \forall\, j < k+1$$

- Condition to be met for k^{th} hinge
  - Hinge k (connecting bodies k and (k+1)) is free, i.e. its generalized force is 0, i.e.

$$0 = \tau(k) = H(k)\, \mathfrak{f}(k) = H(k)\, P(k)\, \alpha(k)$$

Floppy Articulated body model - Accleration relationship

- if $\alpha(k+1)$ is known then $\mathfrak{f}(k+1)$ and $P(k+1)$ needs to be found
- Acceleration on the inboard side of the hinge rigid body propagation

$$\alpha^{+}(k) = \phi^{T}(k+1, k)\, \alpha(k+1)$$

- And then cross the hinge

$$\alpha(k) = \alpha^{+}(k) + H^{T}(k)\, \ddot{\theta}(k)$$

$$0 = \tau(k) = H(k)\, \mathfrak{f}(k) = H(k)\, P(k)\, \alpha(k)$$
$$= H(k)\, P(k)\, \alpha^{+}(k) + H(k)\, P(k)\, H^{T}(k)\, \theta(\ddot{k})$$

Floppy Articulated body model - Value of $\ddot{\theta}$ induced by $\alpha(k+1)$

- Solving for generalized acceleration at the hinge

$$0 = \tau(k) = H(k)\ P(k)\ \alpha^{+}(k) + H(k)\ P(k)\ H^{T}(k)\ \theta(\ddot{k})$$
$$\Downarrow$$
$$\ddot{\theta} = -\mathcal{D}^{-1}(k)\ H(k)\ P(k)\ \alpha^{+}(k) = -\mathcal{G}^{T}(k)\ \alpha^{+}(k)$$

Where,

$$\mathcal{D}(k) \triangleq H(k)\ P(k)\ H^{T}(k), \qquad \mathcal{G} \triangleq P(k)\ H^{T}(k)\ D^{-1}(k)$$

Floppy Articulated body model - Identity

- From the definitions

$$\mathcal{D}(k) \triangleq H(k)\ P(k)\ H^T(k)$$
$$\mathcal{G} \triangleq P(k)\ H^T(k)\ D^{-1}(k)$$

it follows that,

$$H(k)\ \mathcal{G}(k) = I$$

Floppy Articulated body model - Value of $\alpha(k)$ induced by $\alpha(k+1)$

$$\ddot{\theta} = -\mathcal{G}^T(k)\,\alpha^+(k) \qquad \text{and} \qquad \alpha(k) = \alpha^+(k) + H^T(k)\,\ddot{\theta}(k)$$

$$\Downarrow$$

$$\alpha(k) = (I - H^T(k)\,\mathcal{G}^T)\,\alpha^T(k) = \bar{\tau}^T(k)\,\alpha^+(k)$$

where

$$\tau(k) \triangleq \mathcal{G}(k)\,H(k), \qquad \bar{\tau}(k) \triangleq I - \tau(k) = I - \mathcal{G}(k)\,H(k)$$

- $\bar{\tau}(k)$ is the transformation from inboard to outboard accelerations when the k^{th} link is locked

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Floppy Articulated body model - $\tau(k)$ and $\bar{\tau}(k)$ are projections

- claim: $\tau(k)$ is a projection
- $\tau(k)$ is a projection if any power of $\tau(k)$ gives back $\tau(k)$
- Proof:

$$H(k)\ \mathcal{G}(k) = I \qquad \text{and} \qquad \tau(k) \triangleq \mathcal{G}(k)\ H(k)$$

$$\tau(k)\ \tau(k) = \mathcal{G}(k)\ H(k)\ \mathcal{G}(k)\ H(k) = \mathcal{G}(k)\ H(k) = \tau(k)$$

- Since $\tau(k)$ is a projection it is easy to verify that $\bar{\tau}(k) \triangleq I - \tau(k) = I - \mathcal{G}(k)\ H(k)$ is a projection too

Floppy Articulated body model - $\tau(k)$ and $\bar{\tau}(k)$ projection properties

- Identities

$$\tau(k)\,\mathcal{G}(k) = \mathcal{G}(k)$$
$$\bar{\tau}^T(k)\,H^T(k) = 0$$

Floppy Articulated body model - More $\tau(k)$ projection properties

- Claim

$$\tau(k)\ P(k) = P(k)\ \tau^T(k) = \tau(k)\ P(k)\ \tau^T(k)$$

$$P(k)\ \bar{\tau}^T(k) = \bar{\tau}(k)\ P(k) = \bar{\tau}(k)\ P(k)\ \bar{\tau}^T(k)$$

- Proof of first identity

$$\tau(k) \triangleq \mathcal{G}(k)\ H(k) \qquad \mathcal{G}(k) \triangleq P(k)\ H^T(k)\ \mathcal{D}^{-1}(k)$$

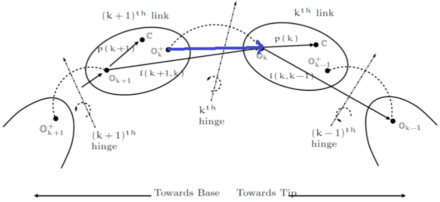
$$\Downarrow$$

$$\tau(k)\ P(k) = P(k)\ H^T(k)\ \mathcal{D}^{-1}(k)\ H(k)\ P(k) = P(k)\ \tau^T(k)$$

also

$$\tau(k)\ P(k) = (\tau(k))^2\ P(k) = \tau(k)\ P(k)\ \tau^T(k)$$

Floppy Articulated body model - Projection of $\alpha^+(k)$



$$\alpha(k) = \alpha^+(k) + H^T(k)\,\ddot{\theta}(k)$$

$$\boxed{\bar{\tau}^T(k)\,H^T(k) = 0}$$

$\bar{\tau}^T(k)$ nullifies hinge accelerations for a floppy hinge

$$\Downarrow$$

$$\boxed{\alpha(k) = \bar{\tau}^T(k)\,\alpha^+(k)}$$

$\bar{\tau}^T(k)$ removes hinge contribution when transmitting accelerations across the floppy hinge

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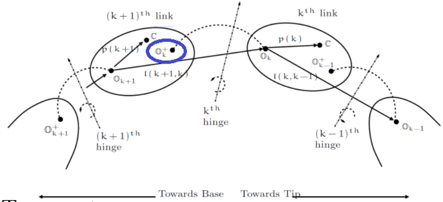
Floppy Articulated body model - Crossing the hinge with $P(k)$



$$\alpha(k) = \bar{\tau}^T(k)\ \alpha^+(k)$$

$$\mathfrak{f}(k) = P(k)\ \alpha(k) = P(k)\ \bar{\tau}^T(k)\ \alpha^+(k) = P^+(k)\ \alpha^+(k)$$

$$\Downarrow$$

$$P^+(k) \triangleq P(k)\ \bar{\tau}^T(k)$$

- ATBI on the inboard side of the hinge

Floppy Articulated body model - $P^+(k)$ Identity

- Have

$$P(k)\,\bar{\tau}^T(k) = \bar{\tau}(k)\,P(k) = \bar{\tau}(k)\,P(k)\,\bar{\tau}^T(k)$$

$$P^+(k) \triangleq P(k)\,\bar{\tau}^T(k)$$

$$\Downarrow$$

$$P^+(k) = \bar{\tau}(k)\,P(k) = P(k)\,\bar{\tau}^T(k) = \bar{\tau}(k)\,P(k)\,\bar{\tau}^T(k)$$

Floppy Articulated body model - $P(k+1)$ ATBI Expression

- Claim:

$$P(k+1) \triangleq \phi(k+1,k)\ P^{+}(k)\ \phi^{T}(k+1,k) + M(k+1)$$

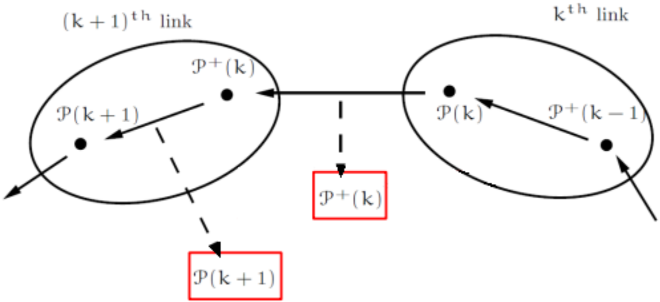
- Proof:

$$\alpha^{+}(k) = \phi^{T}(k+1,k)\ \alpha(k+1) \qquad \text{and} \qquad \mathfrak{f}(k) = P^{+}(k)\ \alpha^{+}(k)$$

- Force balance for the $(k+1)^{th}$ body

$$\begin{aligned}
\mathfrak{f}(k+1) &= \phi(k+1,k)\ \mathfrak{f}(k) + M(k+1)\ \alpha(k+1) \\
&= \phi(k+1,k)\ P^{+}(k)\ \alpha^{+}(k) + M(k+1)\ \alpha(k+1) \\
&= \big(\phi(k+1,k)\ P^{+}(k)\ \phi^{T}(k+1,k) + M(k+1)\big)\ \alpha(k+1) \\
\mathfrak{f}(k+1) &= P(k+1)\ \alpha(k+1)
\end{aligned}$$

Floppy Articulated body model - Floppy decomposition summary



Defining $\phi(k+1,k)$

- Define the articulated body transformation matrix

$$\psi(k+1,k) \triangleq \phi(k+1,k)\,\bar{\tau}(k)$$

  - $\psi(k+1,k)$ is a 6x6 matrix like $\phi(k+1,k)$
  - However it is typically singular as the projection $\bar{\tau}(k)$ is singular
  - It depends on hinge properties through $\bar{\tau}(k)$
  - Unlike $\phi(k+1,k)$ which propagates across rigid bodies, $\psi(k+1,k)$ propagates across articulated bodies

ATBI Riccati Equation

- Claim:

$$P(k+1) = \psi(k+1,k)\ P(k)\ \psi^T(k+1,k) + M(k+1)$$

- Proof:

$$P(k+1) \triangleq \phi(k+1,k)\ P^+(k)\ \phi^T(k+1,k) + M(k+1)$$

The result follows from substituting in,

$$P^+(k) \triangleq P(k)\ \bar{\tau}^T(k)$$

$$\psi(k+1,k) \triangleq \phi(k+1,k)\ \bar{\tau}^T(k)$$

Riccati vs Lyapunov Equations

- Lyapunov equation for CRBs

$$R(k) = \phi(k, k-1)\ R(k-1)\ \phi^T(k, k-1) + M(k)$$

- Riccati equation for ATBI

$$P(k+1) = \psi(k+1, k)\ P(k)\ \psi^T(k+1, k) + M(k+1)$$

- They look similar, so why the different terminology?
- Depends on $P(k)$ and it can be shown that this dependency is quadratic instead of linear as was the case for the Lyapunov equation

O(N) recursive gather algorithm for ATBIs

- This is a tip to base gather recursion

$$P^{+}(0) = 0$$

for $k \quad 1 \ldots n$

$$P(k) = \phi(k, k-1)\ P^{+}(k-1)\ \phi^{T}(k, k-1) + M(k)$$

$$\mathcal{D}(k) = H(k)\ P(k)\ H^{T}(k)$$

$$\mathcal{G}(k) = P(k)\ H^{T}(k)\ \mathcal{D}^{-1}(k)$$

$$\bar{\tau}(k) = I - \mathcal{G}(k)\ H(k)$$

$$P^{+}(k) = \bar{\tau}(k)\ P(k)$$

end loop

Properties of the Articulated Body Inertia

$P(k)$ os not a spatial inertia

- The articulated body inertia $P(k)$ acts like an inertia but is not a spatial inertia!
  - It is a dense 6x6, symmetric, positive definite matrix

Comparison with $P^+(k)$

$$P^+(k) \triangleq P(k)\ \bar{\tau}^T(k)$$

- $P^+(k)$ symmetric, but singular and only positive semi-definite
- Moreover

$$P(k) \geq P^+(k)$$

Comparison with other inertias

$$P^{+}(k) \triangleq P(k)\,\bar{\tau}^{T}(k)$$

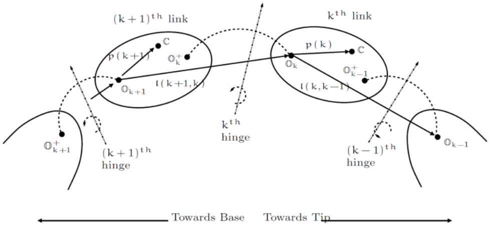
- Also

$$R(k) \geq P(k) \geq M(k)$$

- Composite body inertia is all bodies if outboard bodies were locked
- Articulated body inertia is all bodies if outboard bodies were floppy

ATBI inertia for hinge special cases

Special case: Locked hinge (0 dof)



- Parent/child bodies rigidly coupled (no articulation)
  - $\mathcal{D}(k) = 0, \qquad \mathcal{G}(k) = 0, \qquad \tau(k) = 0$
  - $\bar{\tau}(k) = I$, so all acceleration goes across the locked hinge
  - $P(k) = P^+(k)$
-

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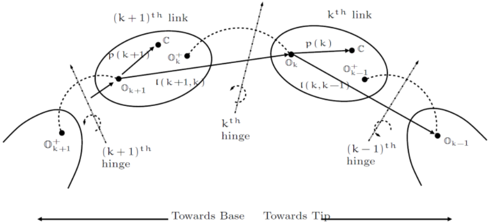
Special case: Uncoupled hinge (6 dof)



- Parent/child bodies uncoupled (no constraints)
  - $\mathcal{D}(k) = P(k) = \mathcal{G}(k)$
  - $\tau(k) = I$
  - $\bar{\tau}(k) = 0, \qquad P^+(k) = 0$, northing carries across the hinge since the are uncoupled

Non-Floppy Articulated body model

Allowing non-zero generalized forces

- For the floppy model, we assumed that the outboard generalized forces were zero and had

$$\mathfrak{f}(k) = P(k)\,\alpha(k) \quad \text{where} \quad \tau(j) = H(j)\,\mathfrak{f}(j) = 0 \quad \forall\, j < k$$

- What happens when the outboard generalized forces are non-zero?
- Look for decomposition of the form:

$$\mathfrak{f}(k+1) = P(k+1)\,\alpha(k+1) + \underbrace{\boxed{\xi(k+1)}}_{\text{residual term}}$$

- Residual to compensate for non-zero gen forces

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Tip body's ATBI decomposition

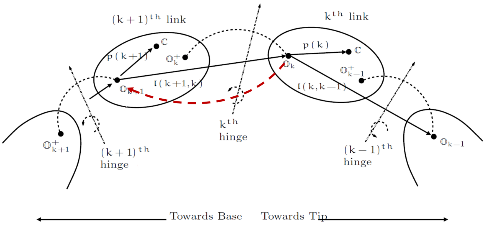
- Similar to before the tip body has no children so,

$$\mathfrak{f}(1) = P(1)\,\alpha(1)$$
$$\xi(1) = 0$$

- Will once again use induction based argument to extend to other bodies
- So let us assume we have established the decomposition for the k^{th} body

$$\mathfrak{f}(k) = P(k)\,\alpha(k) + \xi(k)$$

Induction based derivation - start



- Assume we know the relationship for the k^{th} body:

$$\mathfrak{f}(k) = P(k)\,\alpha(k) + \xi(k)$$

- Want to establish the relationship for the $(k+1)^{th}$ body:

$$\mathfrak{f}(k+1) = P(k+1)\,\alpha(k+1) + \xi(k+1)$$

Moving to the inboard side of the hinge

- Claim:

$$\mathfrak{f}(k) = P^+(k)\ \alpha^+(k) + \xi^+(k) \quad \text{where} \quad \xi^+(k) \triangleq \bar{\tau}(k)\ \xi(k) + \mathcal{G}(k)\ \tau(k)$$

- Proof:

$$\begin{aligned}
\mathfrak{f}(k) &= \bar{\tau}(k)\ \mathfrak{f}(k) + \tau(k)\ \mathfrak{f}(k) = \mathcal{G}(k)\ H(k)\ \mathfrak{f}(k) + \bar{\tau}(k)(P(k)\ \alpha(k) + \xi(k)) \\
&= \mathcal{G}(k)\tau(k) + \bar{\tau}(k)\ P(k)\ \alpha(k) + \bar{\tau}(k)\ \xi(k) \\
&= \mathcal{G}(k)\ \tau(k) + P^+(k)(\alpha^+(k) + H^T(k)\ \ddot{\theta}(k)) + \bar{\tau}(k)\ \xi(k) \\
&= \mathcal{G}(k)\ \tau(k) + P^+(k)\ \alpha^+(k) + \bar{\tau}(k)\ \xi(k)
\end{aligned}$$

Innovation term $\epsilon(k)$

- Have:

$$\xi^{+}(k) \triangleq \bar{\tau}(k)\ \xi(k) + \mathcal{G}(k)\ \tau(k)$$

- Define:

$$\boxed{\epsilon(k) \triangleq \tau(k) - H(k)\ \xi(k)}, \qquad \nu(k) \triangleq \mathcal{D}^{-1}(k)\ \epsilon(k)$$

Innovation term

$$\xi^{+}(k) \triangleq \xi(k) + \mathcal{G}(k)\ \epsilon(k)$$

Moving to $(k+1)$ body frame

- Claim:

$$\mathfrak{f}(k+1) = P(k+1)\,\alpha(k+1) + \xi(k+1)$$

where

$$\xi(k+1) \triangleq \phi(k+1)\,\xi^{+}(k) = \psi(k+1,k)\,\xi(k) + \mathcal{K}(k+1,k)\,\tau(k)$$

with

$$\mathcal{K}(k+1,k) \triangleq \phi(k+1,k)\,\mathcal{G}(k)$$

Moving to $(k+1)$ body frame

- Proof:

$$\begin{aligned}
\mathfrak{f}(k+1) &= \phi(k+1,k)\,\mathfrak{f}(k) + M(k+1)\,\alpha(k+1) \\
&= \phi(k+1,k)(P^{+}(k)\,\alpha^{+}(k) + \xi^{+}(k)) + M(k+1)\,\alpha(k+1) \\
&= \phi(k+1,k)\,P^{+}(k)\,\phi^{T}(k+1,k)\,\alpha^{+}(k) + M(k+1)\alpha(k+1) \\
&+ \phi(k+1,k)\,\xi^{+}(k) \\
&= P(k+1)\,\alpha(k+1) + \phi(k+1,k)\,\xi^{+}(k)
\end{aligned}$$

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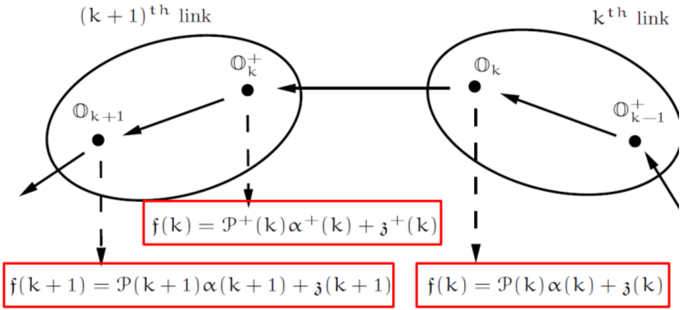
ATBI decomposition summary



$O(N)$ recursive gather algorithm for the residual forces

- Tip-to-base gather recursion

$$\epsilon^+(0) = 0$$

for $k \quad 1 \ldots n$

$$\xi(k) = \phi(k, k-1)\, \xi^+(k-1)$$

$$\epsilon(k) = \tau(k) - H(k)\, \xi(k)$$

$$\xi^+(k) = \xi(k) + \mathcal{G}(k)\, \epsilon(k)$$

end loop

- This, together with earlier recursion for the ATBIs, is half the story for the O(N) forward dynamics algorithm!

Generalized accelerations

- For floppy case we had

$$\ddot{\theta}(k) = -\mathcal{G}^T(k)\ \alpha^+(k)$$

- Claim

$$\ddot{\theta}(k) = \nu(k) - \mathcal{G}^T(k)\ \alpha^+(k)$$

- $\nu(k)$ non-floppiness compensating term
- Proof:

$$\begin{aligned}\ddot{\theta}(k) &= \mathcal{D}^{-1}(k)(\epsilon(k) - H(k)\ P(k)\ \phi^T(k+1,k)\ \alpha(k+1)) \\ &= \nu(k) - \mathcal{G}^T(k)\ \phi(k+1,k)\ \alpha(k+1) = \nu(k) - \mathcal{K}^T(k+1,k)\ \alpha(k+1)\end{aligned}$$

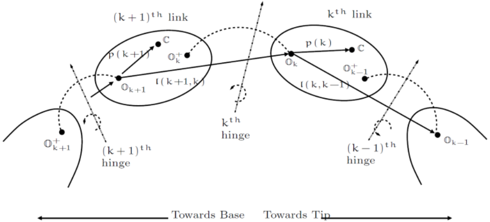
Body spatial accelerations

- Generally have $\alpha(k) = \phi^T(k+1,k)\,\alpha(k+1) + H^T(k)\,\ddot{\theta}(k)$
- Claim $\alpha(k) = \psi^T(k+1,k)\,\alpha(k+1) + H^T(k)\,\nu(k)$
- $\psi^T(k+1,k)$ articulated body transformation matrix
- $H^T(k)\,\nu(k)$ non-floppiness compensating term
- Proof:

$$\begin{aligned}
\alpha(k) &= \phi^T(k+1,k)\,\alpha(k+1) + H^T(k)\,\ddot{\theta}(k)\\
&= \phi^T(k+1,k)\,\alpha(k+1) + H^T(k)(\nu(k) - \mathcal{K}^T(k+1,k)\,\alpha(k+1))\\
&= \psi^T(k+1,k)\,\alpha(k+1) + H^T(k)\,\nu(k)
\end{aligned}$$

ATBI residual for hinge special cases

Special case: Locked hinge (0 dof)



- Parent/child bodies rigidly coupled (no articulation)
  - $\mathcal{D}(k) = 0, \qquad \mathcal{G}(k) = 0, \qquad \tau(k) = 0$
  - $\bar{\tau}(k) = I$
  - $\ddot{\theta}(k) = \nu(k) = \epsilon(k) = 0$
  - $\alpha^+(k) = \alpha(k), \qquad P(k) = P^+(k), \qquad \xi^+(k) = \xi(k)$

Articulated body model

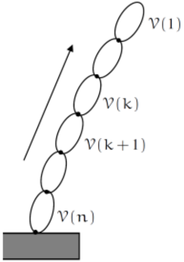
Articulated body model

- An improvement over terminal body model since outboard bodies are not ignored
- Residual term is zero if outboard generalized forces are zero (outboard bodies are floppy)

$$\mathfrak{f}(k) = P(k)\ \alpha(k)$$

- The residual term accounts for the non-zero generalized forces of the outboard bodies

$$\mathfrak{f}(k) = P(k)\ \alpha(k) + \xi(k)$$



Inter-body spatial force decompositions

- Force decompositions consist of inertia + residual terms
- Terminal body model

$$\mathfrak{f}(k) = \underbrace{\boxed{M(k)\ \alpha(k)}}_{k^{th}\text{ body}} + \underbrace{\boxed{\phi(k,k-1)\ \mathfrak{f}(k-1)}}_{\text{all bodies}}$$

- Using CRBs we have

$$\mathfrak{f}(k) = \underbrace{\boxed{R(k)\ \alpha(k)}}_{\text{outboard bodies}} + \underbrace{\boxed{y(k)}}_{\text{outboard generasied acclerations}}$$

- Using ATBI we have

$$\mathfrak{f}(k) = \underbrace{\boxed{P(k)\ \alpha(k)}}_{\text{outboard bodies}} + \underbrace{\boxed{\xi(k)}}_{\text{outboard generalised forces}}$$

Connections to Estimation Theory

The optimal estimation problem

- Consider the noisy, discrete, time-domain dynamical system

$$\begin{aligned} x(k) &= \phi(k, k-1)\, x(k-1) + w(k) \\ \tau(k) &= H(k)\, x(k) \end{aligned}$$

- Estimation problems
  - Optimal filtering: At a given time k, and the past observations $T(1) \dots T(k)$, determine the best estimate $z(k)$ for the state $x(k)$. This is a causal problem.
  - Optimal smoothing: Given all observations – past and future, $T(1) \dots T(n)$, determine the best estimate $f(k)$ for $x(k)$. This is an anti-casual problem.

Role of ϵ_ϕ and ϕ

$$x(k) = \phi(k, k-1)\, x(k-1) + w(k)$$
$$\tau(k) = H(k)\, x(k)$$

- Re-expression of the time domain relationship

$$x = \epsilon_\phi\, x + w$$

- Mapping from the full input vector to the full state vector

$$x = \phi\, w$$

- Mapping from the full state vector to the full output vector

$$T = H\, x$$

Optimal Kalman filter

- The optimal filtering process involves the following steps at each time instant:
  1. Use a Riccati equation to propagate the estimation error covariance and define gains to use
  2. Use the previous state estimate to predict the state at the current time
  3. Extract new information (i.e. the innovations term) from the current observation
  4. Use the innovations term to update the predict to compute the filter state estimate

Correspondence to dynamics

- Will use same notation to show correspondence
  - Estimation error covariance P(k)
  - Riccati equation

$$P(k) = \psi(k, k-1)\ P(k-1)\ \psi^T(k, k-1) + M(k)$$

  - Predict step

$$\xi(k) \triangleq \phi(k, k-1)\ \epsilon^+(k-1)$$

  - Update step

$$\boxed{\epsilon(k) \triangleq \tau(k) - H(k)\ \epsilon(k)}, \qquad \boxed{\epsilon^+(k) = \epsilon(k) + \mathcal{G}(k)\ \epsilon(k)}$$

Innovation term — optimal filter estimate

Optimal Kalman smoother

- Based on Bryson-Frazier method
- Uses a recursion going backwards in time
- Uses the stored optimal filter estimates
- Update the filter estimates using the backward recursion co-state to compute the smoothed state estimate

$$\alpha(k) = \psi(k+1,k)\ \alpha(k+1) + H^T(k)\ \nu(k)$$
$$\mathfrak{f}(k) = \xi(k) + P(k)\ \alpha(k) = \xi^+(k) + P^+(k)\ \alpha^+(k)$$

Key covariance quantities

- State covariance, $cov(x) = \phi \ M \ \phi^T$
- Output covariance, $cov(T) = M$
- Filter estimate covariance, $cov(z) = \phi K \mathcal{D}^{-1} K^T \phi^T = \tilde{\phi} \tau P \tau^T \tilde{\phi}^T = \phi M \phi^T - (P + \tilde{\phi} P + P \tilde{\phi}^T) = (R - P) + \tilde{\phi}(R - P) + (R - P)\tilde{\phi}^T$
- innovations covariance, $cov(\epsilon) = D$
- innovations alt covariance, $cov(\nu) = \mathcal{D}^{-1}$
- filter error covariance, $cov(e_z) = \psi M \psi^T = P + \tilde{\psi} P + P \tilde{\psi}^T$
- Kalman gain, $\mathcal{G}(k) \triangleq P(k) \ H^T(k) \ \mathcal{D}^{-1}(k)$

Summary

- Developed articulated body model for the decomposition of forces
  - Defined articulated body inertias and related quantities
  - Derived expression for residual forces
  - Developed $O(N)$ gather algorithm for computing these quantities
- Described parallels with estimation theory